

# **Siddhant College of Management Studies**



A Project Report

On

## ***Pathology Lab System***

Submitted in partial fulfillment

Of

Bachelor of Computer Applications

### **SUBMITTED BY**

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*Seat No: 38404*

**Academic Year**

**2017-2018**

# Siddhant College of Management Studies



A Project Report

On

## ***Pathology Lab System***

### **CERTIFICATE**

This is to certify that **Miss. Dhongade Shruti Shashikant** and **Miss. Navi Sujata Basavaraj** of class T.Y.BCAs Satisfactorily completed the Project entitled **Pathology Lab System** in BCA Sem VI Lab Course during the year 2017-2018 as per requirements of the Savitribai Phule Pune University Pune .

**Project Guide**

**Prof. Yogesh Patil**

**Principal**

**Dr. Sagar Manjare**

**Internal Examiner**\_\_\_\_\_

**External Examiner**\_\_\_\_\_

**Date:-**

## **ACKNOWLEDGEMENT**

*I would like to express my gratitude towards a number of people whose support and consideration has been an invaluable asset during the course of this project.*

*First and foremost, I would like to thank my guide **Prof .Yogesh Patil** for providing me with their invaluable guidance throughout the course of this project. It would have been an almost impossible task to complete this project without his support, motivation, valuable suggestions and criticism.*

*I convey my gratitude to our respected Principal **Dr.Sagar Manjare** for their motivation, guidance, and criticism and also for providing various facilities, which helped me greatly in the course of this project.*

*And, last but not the least we would like to thank all the teaching, non-teaching staff of Computer Application Department and also our friends for directly or indirectly helping me for the project completion and all the resources provided.*

**Miss Shruti Dhongade**

**&**

**Miss Sujata Navi**

## DECLARATION

I am **Miss. Dhongade Shruti Shashikant** and **Miss. Navi Sujata Basavaraj** here by declare that this project work entitled **Pathology Lab System** submitted at Siddhant Collage of Management Studies, Sudumbare, Pune, (Affiliated to University of Pune) is a record of original work done by me under the supervision and guidance of **Prof. Yogesh Patil** Department of Computer Application.

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# ***Pathology Lab System***



# INTRODUCTION

The name of the system is ***Pathology Lab System***. It is mainly designed for transaction and billing solution for Pathology tests. It is the user friendly system. Easy maintenance of Pathology database. In this system user is able to add new Customer and report generation, automatically bill generation. It can generate reports such as Monthly, Half Monthly, weekly and Daily Report.

This system is developed in “JAVA” which is used for user interface, it also platform independent system, it can easily run on Linux Other & operating system. & File System is used for storing records. All the information about transaction process is studied by the manual system. This system is user-friendlier than that of the manual system i.e. currently running in the system.



# System Analysis



The face of system analysis includes:

1. Getting familiar with the current working system.
2. Getting information from the agent.
3. Getting familiarized with the workflow with respect to authorities and responsibilities.

System analysis usually includes

- ❖ Fact Findings
- ❖ System Boundaries

- **Fact Finding Technique:-**

A key part of feasibility is gathering information about the present system. The analyst knows what information to gather to make of it.

## **1) Interviews :-**

Analysts use interview to collect information from individuals who they consider should be the sources, who are current users of the existing system. The analyst should have a face conversation with the users & administrator of the system & fixed set of questions is prepared.

## **2) Questionnaires :-**

It allows analyst to collect information from a group of individuals who may or may not be using the system thus resulting some times in irrelevant data & data redundancy.

### **3) Record Review :-**

Consisting of analyzing the previous operations in the company & forecasting the new future schemes .Record include table name, date& time creation, user login etc.

### ***Feasibility Study:-***

***“Feasibility Study”*** is a test of the system according to its workability, impact of the organization, ability to meet user needs and effective use of the resources.

The feasibility study is carried out to test whether the proposed system is worth being implemented. Feasibility study is a test of system proposed regarding its work ability, its impact on the organization ability to meet user needs and effective use of resources. It is usually carried out by a small number of people who are familiar with the information system techniques, understand the part of the business or organization that will be involved or effected by the project and are skilled in the system analysis and design process.

Collection of required data and analysis of proposed system is termed as feasibility study. While considering feasibility study of any system, the system must meet its best performance by its 5 type of feasibility study i.e.

1. Operational Feasibility
2. Technical Feasibility
3. Economical Feasibility
4. Schedule Feasibility
5. Behavioral Feasibility

- **Operational feasibility:**

The study deals with the following issues. Is there sufficient support for the project from the management of exam department & user has been involved in planning & development of project. When the proposed system was tested for above mention issues. It was found that result was tending towards being positive & leading the system to make operational feasible.

After giving very small training to staff members, they can easily operate and familiar with the software.

So, that's why it is operationally feasible to implement the new system.

In this feasibility we consider following points:

1. What changes will be brought with the system.
2. What new skills will be required? Do the existing staff members have these skills?

If not, can they be trained in due course of time?

In the new system we made some major changes for the staff members so that they have to be trained to use the newly added facilities. These major changes are possible and give a new era in the Supermarket in production and sales management.

- **Technical Feasibility:**

A study of resources availability that may affect the ability to achieve an acceptable system. This evaluation determines whether the technology needed for the proposed system is available or not. This system can be made in any language that support good user interface and easy database handling. Technical needs may include:

- **Economical Feasibility:-**

The paper work is reduced & also the data storage cost will be reduced as the data backup of one complete program can be taken on a 1.44 MB floppy disk. So the new system to be work out cost efficient system. As the system is fully automated it requires very less time for the entry of transactions.

- **Schedule Feasibility:-**

Time evaluation is most important consideration in development of the project. So the project is concerned should be completed with fixed in scheduled time as far as company is concerned. New system is not so much big so it is easy to make in few days.

## ***SOFTWARE REQUIRMENTS***

### **HARDWARE USED:-**

A Computer with following minimum configurations to implement the system.

- PC Pentium IV
- 512 MB RAM
- 1.44 MB Floppy drive
- 1 TB Hard disk
- 15 Inch Color Monitor
- Keyboard
- Printer
- Uninterruptible power Supply

## **Software Used:-**

- Front End: Java
- Back End: Microsoft Access 2013

## **Platform Support:**

- Microsoft windows 8.1
- Microsoft windows 2013

## Need for Computerization:-

Today almost every hospitals, etc. are providing pathological test facilities to the public. But their **PATHOLOGY LAB SYSTEM** is manual which has lots of drawbacks.

Today almost every system is computerized and these computerized system are always have lot of advantages over manual system. According to my survey no one has design a computerized system for **PATHOLOGY LAB SYSTEM**. If we computerized **PATHOLOGY LAB SYSTEM** that will remove all drawbacks of the manual system. The main aim of the computerized system is to provide a proper management of Patients Details, Test Details, Admin /Doctors Details and eliminate the drawback of the manual system.

## **Existing System:-**

Existing System is manual. Existing system uses register books at both gates for the entry of the patients with their ID, Name, Date, Time and current Type of Test.

### **Advantages of Existing System**

- As the system is manual there is no need of power.
- As information is stored in register books hard copy of record is present.

### **Disadvantages of Existing System**

- We have to maintain information in register books.
- The Existing system is time consuming.
- More human errors.
- More manpower require to maintain the system.
- Searching a record is a tedious task
- Wastage of paper & stationeries.
- Take more time for generating result.
- It is costly system.
- Possibility of loss of data.

## **Proposed System:-**

Proposed system **PATHOLOGY LAB SYSTEM** is fully computerized system removes all the drawbacks of the existing system providing lot of facilities for user (Administrator) such as allocating patient details and test details.

### ***Objectives of Proposed System***

- To reduce manpower.
- To reduce manual work.
- To provide a user friendly system.
- To store large volume of correct and accurate information.
- To provide backup facility.
- To facilitate speedy access to any particular details.

### ***Features of Proposed System***

- Reduced manpower for system.
- Human Errors are reduced.
- Working is fast i.e time saved.
- Searching is Fast.
- Printing Roaster for Driver.
- Reduced paper work and stationary.
- Ex- Drivers details are also maintained.
- Drivers and Employees details are maintained and managed.



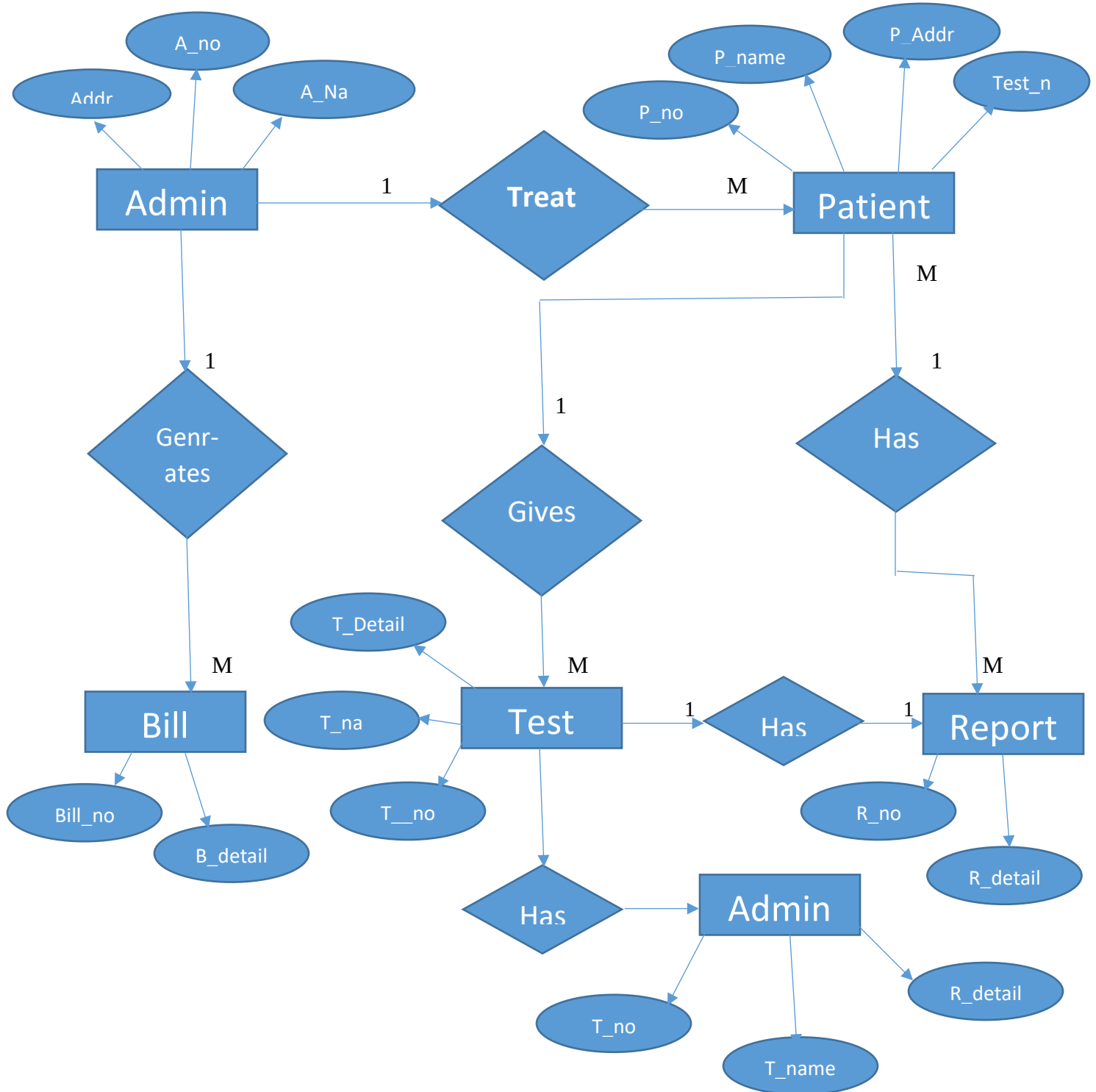
- System is more secure(only Administrator can access the system).
- Searching in Log Register is possible on the basis of

Patient Id, Patient name.

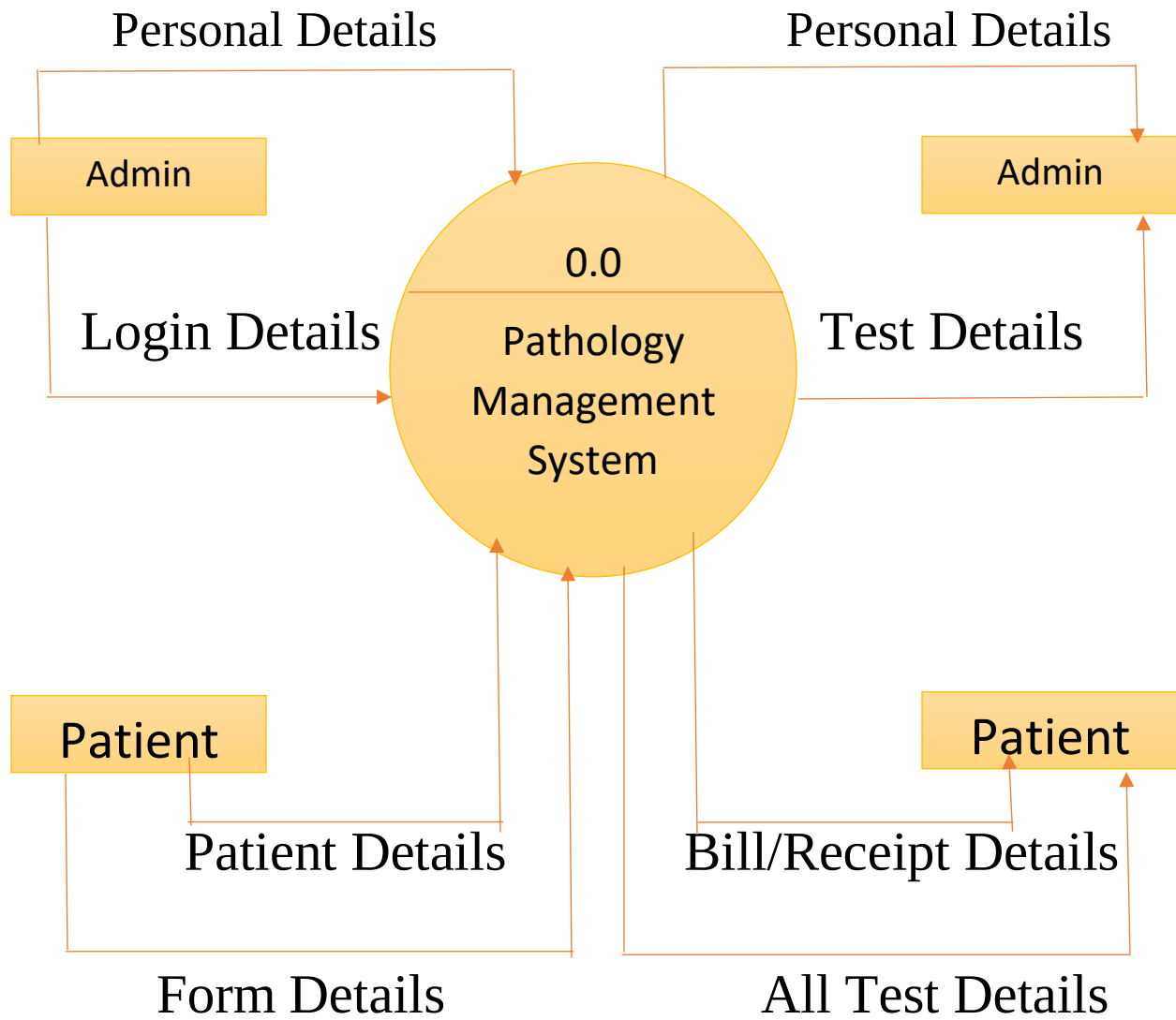
- Reports are generated quickly.
- Data entered is cross-validated.

# System Design

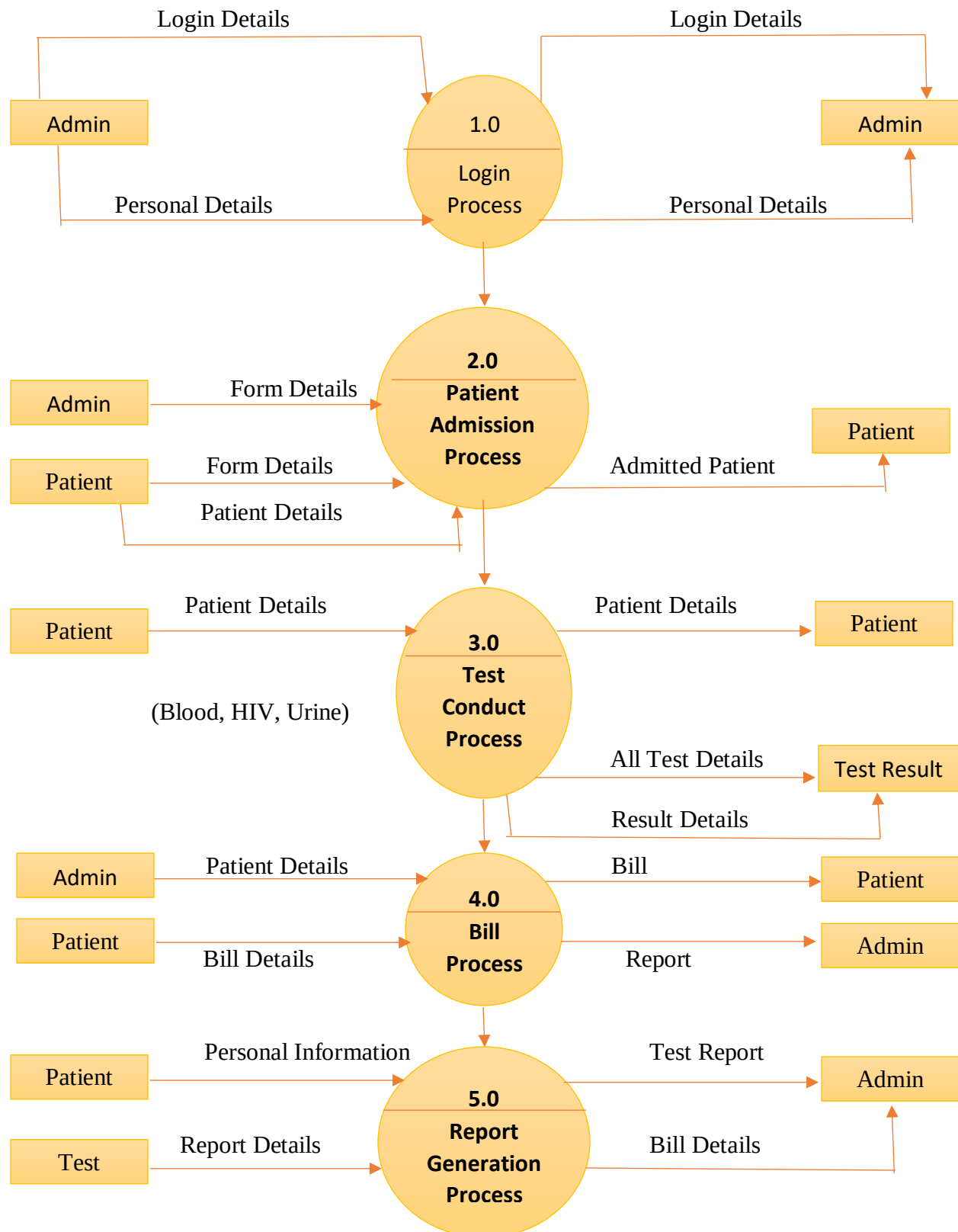
## *E-R Diagram:-*



### ***Context Level DFD:-***



## First Level DFD:-



## **Data Dictionary:-**

### ➤ **Login Information:-**

| <b>Field Name</b> | <b>Field Datatype</b> | <b>Description</b> |
|-------------------|-----------------------|--------------------|
| User Name         | Text                  | User Name          |
| Password          | Text                  | Password           |

### ➤ **Patient Admission Form:-**

| <b>Field Name</b> | <b>Field Datatype</b> | <b>Description</b>  |
|-------------------|-----------------------|---------------------|
| Regd No.          | Number                | Register Number     |
| ADMIN_TYPE        | Text                  | OPD Or IPD          |
| FN                | Text                  | Patient First Name  |
| MN                | Text                  | Patient Middle Name |
| LN                | Text                  | Patient Last Name   |
| AGE               | Text                  | Patient Age         |
| ADDR              | Text                  | Patient Address     |
| SEX               | Text                  | Patient Sex         |
| REF               | Text                  | Doctor Reference    |

➤ **Bill Details:-**

| <b>Field Name</b> | <b>Field Datatype</b> | <b>Description</b>              |
|-------------------|-----------------------|---------------------------------|
| Regdno            | Number                | Register Number                 |
| Rupees            | Number                | Amount                          |
| Received          | Text                  | Amount Received From            |
| Heamo             | Number                | Heamogram Test Amount           |
| Blood             | Number                | Blood Test Amount               |
| Bloch             | Number                | Blood Chemistry Amount          |
| Urine             | Number                | Urine Test Amount               |
| Thyroid           | Number                | Thyroid Test Amount             |
| Bsugar            | Number                | Blood Sugar Fasting Test Amount |
| Hiv               | Number                | HIV Test Amount                 |

## ➤ *Heamogram Test:-*

| <b>Field Name</b> | <b>Field Datatype</b> | <b>Description</b>       |
|-------------------|-----------------------|--------------------------|
| Heamogram         | Number                | Heamogram Details        |
| RBCCount          | Number                | RBCCount Details         |
| WBCCount          | Number                | WBCCount Details         |
| Platelets         | Number                | Platelets Details        |
| PackedCellVolume  | Number                | PackedCellVolume Details |
| MCV               | Number                | MCV Details              |
| MCHCount          | Number                | MCHCount Details         |
| MCHCCount         | Number                | MCHCCount Details        |
| Neutrophills      | Number                | Neutrophills Details     |
| Lymphocytes       | Number                | Lymphocytes Details      |
| Esinophills       | Number                | Esinophills Details      |
| Monocytes         | Number                | Monocytes Details        |
| Basophills        | Number                | Basophills Details       |
| PrematureCells    | Text                  | PrematureCells Details   |
| BloodParasites    | Text                  | BloodParasites Details   |
| Comment           | Text                  | Comment For Test         |

### ➤ **Blood Chemistry:-**

| <b>Field Name</b> | <b>Field Datatype</b> | <b><i>1Description</i></b> |
|-------------------|-----------------------|----------------------------|
| BloodUreaLevel    | Number                | Blood Lavel Details        |
| Creatine          | Number                | Creatine Details           |
| Sodium            | Number                | Sodium Details             |
| Potassium         | Number                | Potassium Details          |
| Chloride          | Number                | Chloride Details           |
| Comment           | Text                  | Comment For This Test      |
| Dr                | Text                  | Doctor Details             |

### ➤ **Blood Group:-**

| <b>Field Name</b> | <b>Field Datatype</b> | <b><i>Description</i></b> |
|-------------------|-----------------------|---------------------------|
| Bgroup            | Text                  | Blood Group details       |
| Comment           | Text                  | Commment For This Test    |
| Dr                | Text                  | Doctor Details            |



➤ ***Blood Sugar Fasting Test :-***

| <b>Field Name</b> | <b>Field Datatype</b> | <b><i>Description</i></b> |
|-------------------|-----------------------|---------------------------|
| RegdNo            | Number                | Registration Number       |
| BloodSugarLevel   | Number                | Blood Sugar Level Details |
| UrineSugar        | Text                  | Urine Sugar Details       |
| UrineAcetone      | Text                  | Urine Acetone Details     |
| Comment           | Text                  | Comment For This Test     |

➤ ***HIV Test:-***

| <b>Field Name</b> | <b>Field Datatype</b> | <b><i>Description</i></b> |
|-------------------|-----------------------|---------------------------|
| Result            | Text                  | Result Details            |
| MethodUsed        | Text                  | Which Method Used         |
| Comment           | Text                  | Comment For This Test     |
| DR                | Text                  | Doctor Details            |

## Report Details:-

- **Receipt Details**

| <b>Field Name</b> | <b>Field Datatype</b> | <b>Description</b>           |
|-------------------|-----------------------|------------------------------|
| Regdno            | Number                | Registration Number          |
| Rno               | Number                | Receipt Number               |
| Pmode             | Test                  | Cash,Cheque,Credit Card.     |
| ReceiptAmount     | Number                | Receipt Amount Details       |
| TotalReceipt      | Number                | Total Receipt Amount Details |
| PreparedBy        | Test                  | Prepared By Person Sign.     |
| CheckedBy         | Test                  | Checked By Person Sign.      |
| AuthorisedBy      | Test                  | Authorised By Person Sign.   |

- **Urine Test :-**

| <b>Field Name</b> | <b>Field Datatype</b> | <b>Description</b> |
|-------------------|-----------------------|--------------------|
| Quantity          | Number                | Quantity           |
| Appearance        | Text                  | Appearance Details |
| Colour            | Text                  | Colour t Details   |
| Reaction          | Text                  | Reaction Details   |

|              |      |                       |
|--------------|------|-----------------------|
| SpecificGra  | Text | SpecificGra Details   |
| BloodComment | Text | BloodComment Details  |
| Leucocytes   | Text | Leucocytes Details    |
| RedBcell     | Text | RedBcell Details      |
| Epicell      | Text | Epicell Details       |
| Protein      | Text | Protein Details       |
| Gluko        | Text | Gluko Details         |
| Bilesal      | Text | Bilesal Details       |
| Bilepig      | Text | Bilepig Details       |
| Keto         | Text | Keto Details          |
| Uroboli      | Text | Uroboli Details       |
| Casts        | Text | Casts Details         |
| Miscel       | Text | miscel Details        |
| Comt         | Text | Comment For This Test |
| Dr           | Text | Doctor Details        |

• ***Heamogram Test:-***

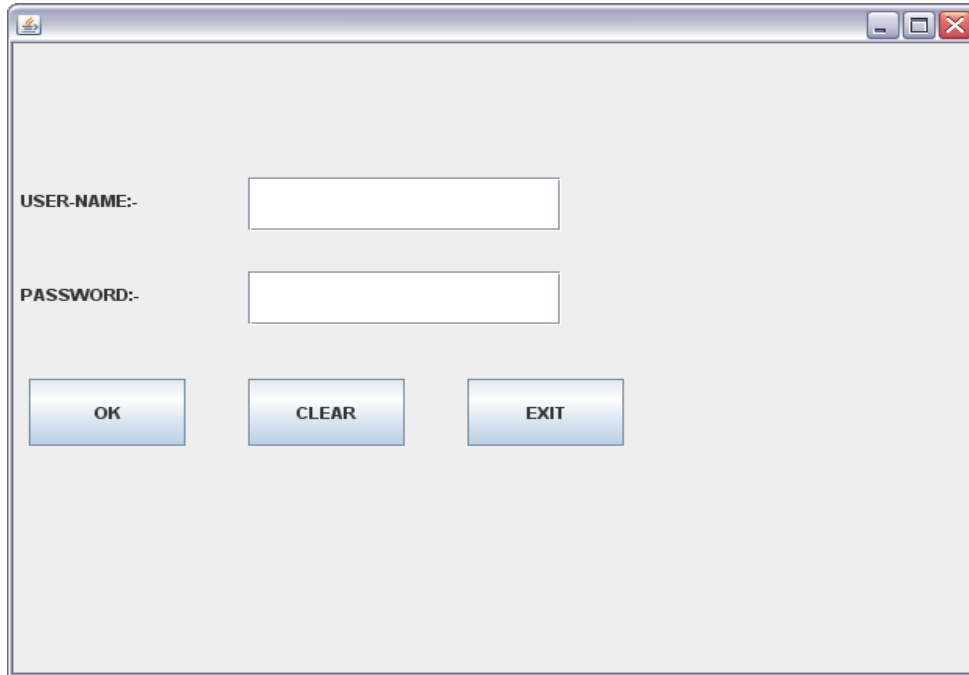
| <b>Field Name</b> | <b>Field Datatype</b> | <b><i>Description</i></b> |
|-------------------|-----------------------|---------------------------|
| Heamogram         | Number                | Heamogram Details         |
| RBCCount          | Number                | RBCCount Details          |
| WBCCount          | Number                | WBCCount Details          |
| Platelets         | Number                | Platelets Details         |
| PackedCellVolume  | Number                | PackedCellVolume Details  |
| MCV               | Number                | MCV Details               |
| MCHCount          | Number                | MCHCount Details          |
| MCHCCount         | Number                | MCHCCount Details         |
| Neutrophills      | Number                | Neutrophills Details      |
| Lymphocytes       | Number                | Lymphocytes Details       |
| Esinophills       | Number                | Esinophills Details       |
| Monocytes         | Number                | Monocytes Details         |
| Basophills        | Number                | Basophills Details        |
| PrematureCells    | Text                  | PrematureCells Details    |
| BloodParasites    | Text                  | BloodParasites Details    |
| Comment           | Text                  | Comment For Test          |

- ***Blood Chemistry:-***

| <b>Field Name</b> | <b>Field Datatype</b> | <b><i>Description</i></b> |
|-------------------|-----------------------|---------------------------|
| BloodUreaLevel    | Number                | Blood Lavel Details       |
| Creatine          | Number                | Creatine Details          |
| Sodium            | Number                | Sodium Details            |
| Potassium         | Number                | Potassium Details         |
| Chloride          | Number                | Chloride Details          |
| Comment           | Text                  | Comment For This Test     |
| Dr                | Text                  | Doctor Details            |

# Form Design

## *Login Form:-*

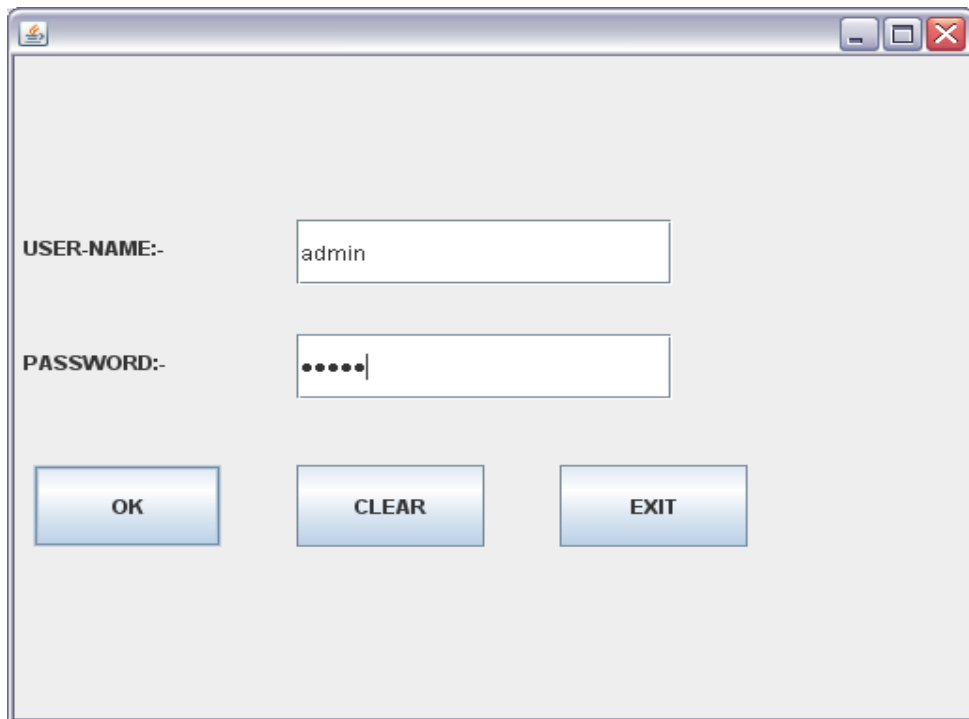


USER-NAME:-

PASSWORD:-

OK CLEAR EXIT

## OUTPUT:-



USER-NAME:-

PASSWORD:-

OK CLEAR EXIT

## Main Form:-

 **PATIENT ADMISSION FORMS**   

### PATHOLOGY TESTS

DR. PARANJPE'S PATHOLOGY  
1098/18, NEAR MODEL COLONY POST OFFICE, SHIVAJINAGAR, PUNE  
PH.NO:-98770005332

Regd. No:  ADMISSION TYPE

FIRST NAME MIDDLE NAME LAST NAME AGE

ADDRESS

SEX

RECIEVED:

REPORT:

REF. DOCT...

### ENTER THE TEST TO BE PERFORMED

|                                                |                                                         |                                                |
|------------------------------------------------|---------------------------------------------------------|------------------------------------------------|
| <input type="button" value="HEAMOGRAM TEST"/>  | <input type="button" value="URINE EXAMINATION"/>        | <input type="button" value="HIV TEST"/>        |
| <input type="button" value="BLOOD TEST"/>      | <input type="button" value="THYROID TEST"/>             | <input type="button" value="RECEIPT DETAILS"/> |
| <input type="button" value="BLOOD CHEMISTRY"/> | <input type="button" value="BLOOD SUGAR FASTING TEST"/> | <input type="button" value="BILL DETAILS"/>    |

## Output:-

 **PATIENT ADMISSION FORMS**

**PATHOLOGY TESTS**  
DR. MAHESH PATHOLOGY  
SHIVKRUPA COLONY,NADHENAGAR,KALEWADI,PIMPRI,PUNE-411017.  
PH.NO:-9730986253

Regd. No:

ADMISSION TYPE

SAVE

FIRST NAME MIDDLE NAME LAST NAME

ADDRESS

AGE

RESET

SEX

EXIT

RECIEVED:

REPORT:

REF. DOCT...

**ENTER THE TEST TO BE PERFORMED**

HEAMOGRAM TEST

URINE EXAMINATION

HIV TEST

BLOOD TEST

THYROID TEST

RECEIPT DETAILS

BLOOD CHEMISTRY

BLOOD SUGAR FASTING TEST

BILL DETAILS



## Input:-

HEAMOGRAM TEST


### HEAMOGRAM TEST

| RegdNo             | PATIENTS VALUE | UNITS | RANGE    |
|--------------------|----------------|-------|----------|
| HEAMOGLOBIN        | Select One     | gm %  | 9.0-16.0 |
| RBC COUNT          | Select one     | cum   | dfnduf   |
| WBC COUNT          | Select One     | cum   | dfnduf   |
| PLATELETS          | Select one     | cum   | dfnduf   |
| PACKED CELL VOL... | Select One     | cum   | dfnduf   |

---

|               |            |     |        |
|---------------|------------|-----|--------|
| BLOOD INDICES |            |     |        |
| MCV           | Select One | cum | dfnduf |
| MCH COUNT     | Select One | cum | dfnduf |
| MCHC COUNT    | Select One | cum | dfnduf |

---

|                       |            |     |        |
|-----------------------|------------|-----|--------|
| DIFFERNCIAL WBC COUNT |            |     |        |
| NEUTROPHILLS          | Select One | cum | dfnduf |
| LYMPHOCYTES           | Select One | cum | dfnduf |
| ESINOPHILLS           | Select One | cum | dfnduf |
| MONOCYTES             | Select One | cum | dfnduf |
| BASOPHILLS            | Select One | cum | dfnduf |
| Premature cells       | Select One |     |        |

---

|                 |            |  |  |
|-----------------|------------|--|--|
| BLOOD PARASITES | Select One |  |  |
| COMMENT         |            |  |  |

SAVE  
RESET  
EXIT  
BACK  
REPORT

DR. MAHESH

## Output:-

HEAMOGRAM TEST



### HEAMOGRAM TEST

| RegdNo                 | PATIENTS VALUE             | UNITS | RANGE    |
|------------------------|----------------------------|-------|----------|
| 1                      |                            |       |          |
| HEAMOGLOBIN            | 12.4                       | gm %  | 9.0-16.0 |
| RBC COUNT              | 123.45                     | cum   | dfhduf   |
| WBC COUNT              | 4.56                       | cum   | dfhduf   |
| PLATELETS              | 3.40                       | cum   | dfhduf   |
| PACKED CELL VOL...     | 0.06                       | cum   | dfhduf   |
| BLOOD INDICES          |                            |       |          |
| MCV                    | 43.5                       | cum   | dfhduf   |
| MCH COUNT              | 3.45                       | cum   | dfhduf   |
| MCHC COUNT             | 46.7                       | cum   | dfhduf   |
| DIFFERNICIAL WBC COUNT |                            |       |          |
| NEUTROPHILLS           | 4.35                       | cum   | dfhduf   |
| LYMPHOCYTES            | 6.57                       | cum   | dfhduf   |
| ESINOPHILLS            | 6.57                       | cum   | dfhduf   |
| MONOCYTES              | 67.8                       | cum   | dfhduf   |
| BASOPHILLS             | 4.5                        | cum   | dfhduf   |
| Premature cells        | Absent                     |       |          |
| BLOOD PARASITES        | Present                    |       |          |
| COMMENT                | Heamogram Is Under Control |       |          |

SAVE
RESET
EXIT
BACK

Message
RECORDS SAVED SUCCEESUFLY
OK

DR. MAHESH

***Input:-***

 URINE TEST





## ROUTINE URINE EXAMINATION



PHYSICAL EXAMINATION DETAILS

RegdNO

QUANTITY

APPEARANCE

COLOUR

REACTION(PH)

SPECIFIC GRAVITY

CHEMICAL EXAMINATION DETAILS

PROTEIN

GLUCOSE

BILE SALTS

BILE PIGMENTS

KETONES

UROBILINOGEN

Blood Comment

MICROSCOPIC EXAMINATION(of Centrifuged Sa...)

Leucocytes

Red Blood Cells

Epithelial Cells

Casts

Miscellaneous

Comment

SAVE

RESET

EXIT

BACK

REPORT

Dr.PARANJPE

## Output:-

 URINE TEST



### ROUTINE URINE EXAMINATION



**PHYSICAL EXAMINATION DETAILS**

RegdNO

|                  |                                          |
|------------------|------------------------------------------|
| QUANTITY         | <input type="text" value="10.4"/>        |
| APPEARANCE       | <input type="text" value="CLEAR"/>       |
| COLOUR           | <input type="text" value="pale yellow"/> |
| REACTION(PH)     | <input type="text" value="ACIDIC"/>      |
| SPECIFIC GRAVITY | <input type="text" value="dfg"/>         |

**CHEMICAL EXAMINATION DETAILS**

|               |                                      |
|---------------|--------------------------------------|
| PROTEIN       | <input type="text" value="Absent"/>  |
| GLUCOSE       | <input type="text" value="Present"/> |
| BILE SALTS    | <input type="text" value="Present"/> |
| BILE PIGMENTS | <input type="text" value="Present"/> |
| KETONES       | <input type="text" value="Absent"/>  |
| UROBILINOGEN  | <input type="text" value="Present"/> |
| Blood Comment | <input type="text" value="df"/>      |

**MICROSCOPIC EXAMINATION(of Centrifuged Sa...)**

|                  |                                          |
|------------------|------------------------------------------|
| Leucocytes       | <input type="text" value="32.23"/>       |
| Red Blood Cells  | <input type="text" value="6.537"/>       |
| Epithelial Cells | <input type="text" value="0.65"/>        |
| Casts            | <input type="text" value="Present"/>     |
| Miscellaneous    | <input type="text" value="Present"/>     |
| Comment          | <input type="text" value="Urine Is OK"/> |

**SAVE**

**RESET**

**EXIT**

**BACK**

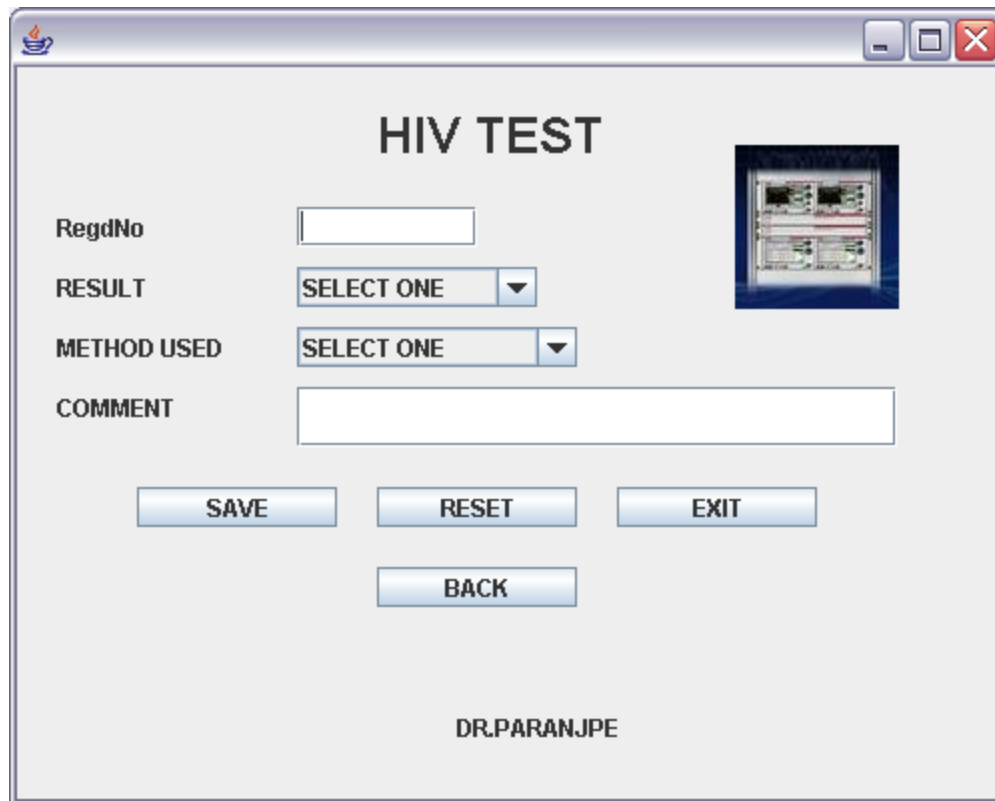
**REPORT**

**Message**

 RECORD INSERTED SUCESSFULLY

**OK**

***Input:-***



The screenshot shows a Windows-style application window titled "HIV TEST". The window has a standard title bar with minimize, maximize, and close buttons. Inside the window, the title "HIV TEST" is centered at the top. To the right of the title is a small image of a circuit board. Below the title, there are four input fields: "RegdNo" (a text box), "RESULT" (a dropdown menu with "SELECT ONE" selected), "METHOD USED" (a dropdown menu with "SELECT ONE" selected), and "COMMENT" (a text box). Below these fields are four buttons: "SAVE", "RESET", "EXIT", and "BACK". At the bottom center of the window, the text "DR.PARANJPE" is displayed.

**HIV TEST**

RegdNo

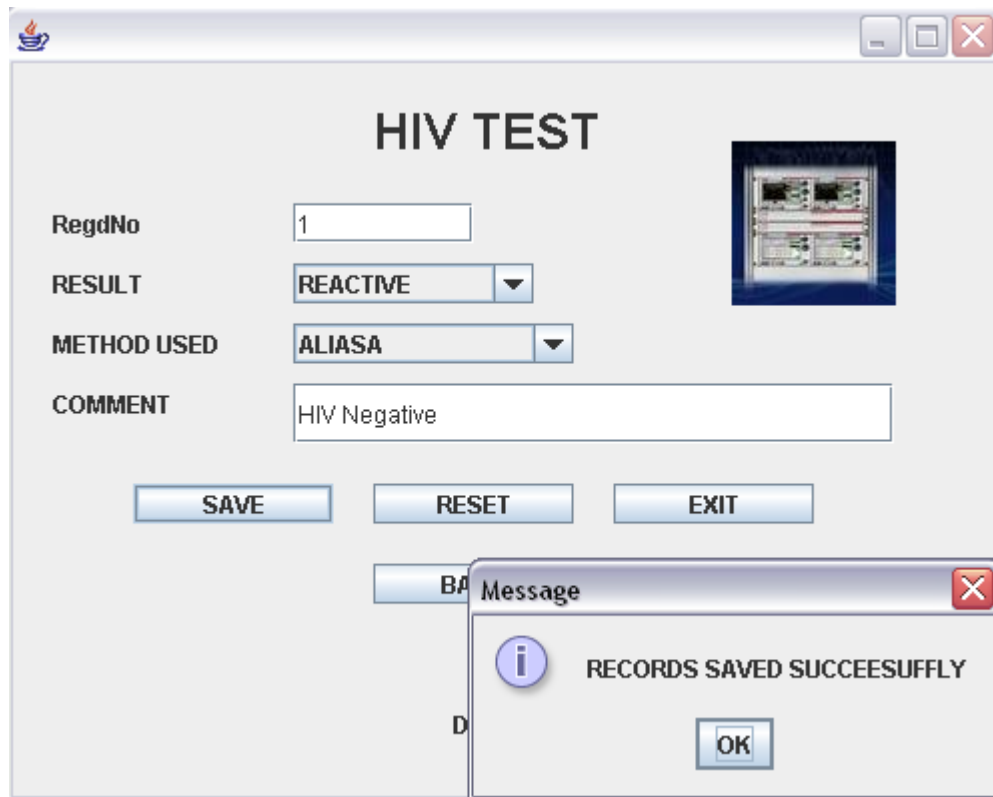
RESULT

METHOD USED

COMMENT

DR.PARANJPE

## Output:-

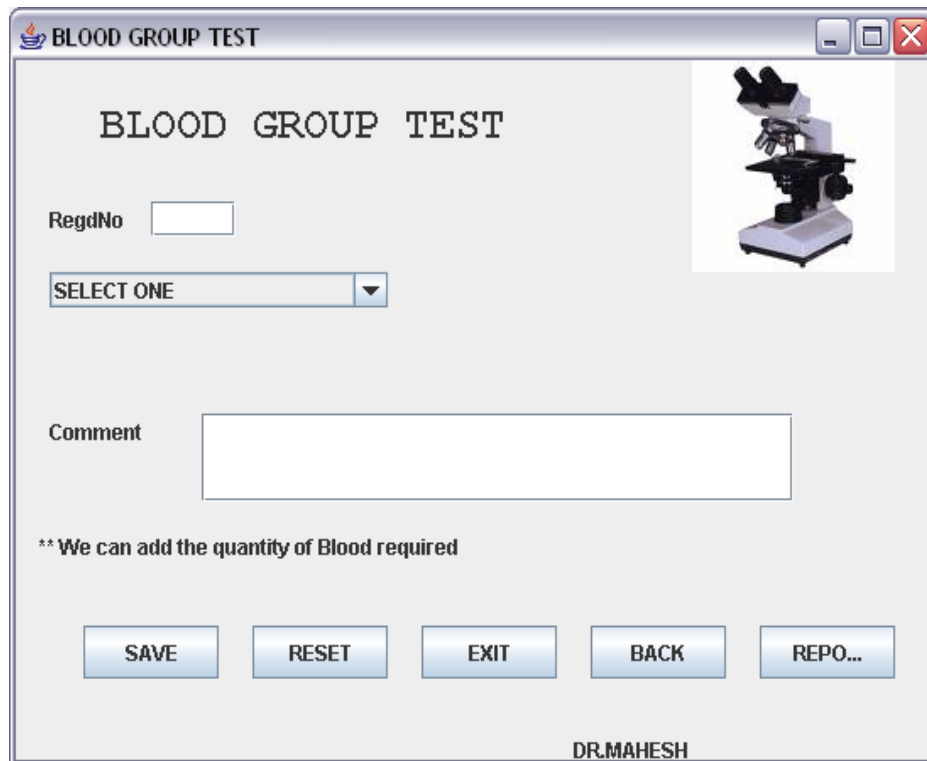


The screenshot displays a Java Swing window titled "HIV TEST". The window contains the following fields and controls:

- RegdNo**: A text input field containing the value "1".
- RESULT**: A dropdown menu currently showing "REACTIVE".
- METHOD USED**: A dropdown menu currently showing "ALIASA".
- COMMENT**: A text input field containing the text "HIV Negative".
- Buttons**: Three buttons labeled "SAVE", "RESET", and "EXIT" are positioned below the input fields.
- Image**: A small image of a computer monitor is located in the top right corner of the window.

A modal message dialog box is open in the foreground, titled "Message". It contains an information icon (i) and the text "RECORDS SAVED SUCCEESUFFLY" (note the spelling error). An "OK" button is at the bottom of the dialog.

***Input:-***



**BLOOD GROUP TEST**

RegdNo

SELECT ONE ▼

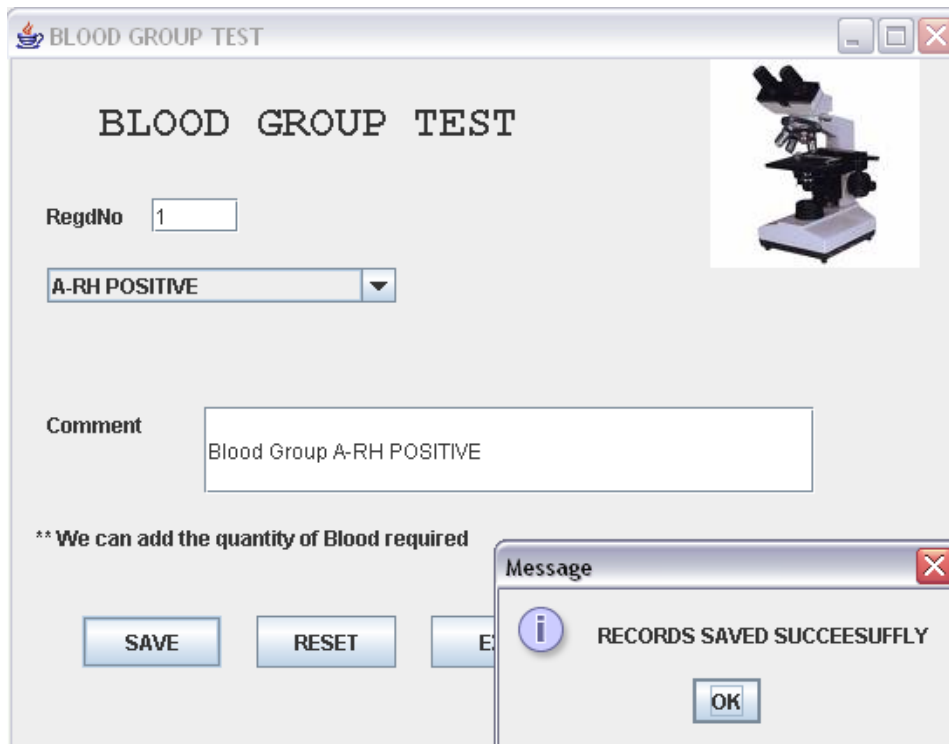
Comment

**\*\* We can add the quantity of Blood required**

SAVE RESET EXIT BACK REPO...

DR.MAHESH

## Output:-



The screenshot displays a software application window titled "BLOOD GROUP TEST". The window contains the following elements:

- Title Bar:** "BLOOD GROUP TEST" with standard minimize, maximize, and close buttons.
- Header:** "BLOOD GROUP TEST" in a large, bold, serif font.
- Form Fields:**
  - RegdNo:** A text input field containing the value "1".
  - Blood Group:** A dropdown menu currently showing "A-RH POSITIVE".
  - Comment:** A text area containing the text "Blood Group A-RH POSITIVE".
- Buttons:** "SAVE", "RESET", and a partially visible "E" button.
- Image:** A small image of a microscope is located in the top right corner of the window.
- Message Box:** A "Message" dialog box is overlaid on the bottom right, displaying an information icon, the text "RECORDS SAVED SUCCEESUFFLY", and an "OK" button.
- Footer:** A note at the bottom left reads: "\*\* We can add the quantity of Blood required".



## Input:-



RegdNo

# THYROID



| TestName                    | Method  | Value                | UNITS  | Range    |
|-----------------------------|---------|----------------------|--------|----------|
| TOTAL_TRIIODOTHYRONINE      | C.L.I.A | <input type="text"/> | ng/dl  | 60-200   |
| TOTAL_THYROXINE             | C.L.I.A | <input type="text"/> | ug/dl  | 4.5-12.0 |
| THYROID_STIMULATING_HORMONE | C.L.I.A | <input type="text"/> | uIU/ml | 0.30-5.5 |

*Technology:*

T3-Competitive Chemi Luminescent Immuno Assay

T4-Competitive Chemi Luminescent Immuno Assay

TSH-Ultra Sensitive Chemi Luminescent Immuno Assay

Dr.Mahesh & ...

## Output:-

RegdNo

# THYROID



| TestName                    | Method  | Value                            | UNITS  | Range    |
|-----------------------------|---------|----------------------------------|--------|----------|
| TOTAL_TRIIODOTHYRONINE      | C.L.I.A | <input type="text" value="98"/>  | ng/dl  | 60-200   |
| TOTAL_THYROXINE             | C.L.I.A | <input type="text" value="5.2"/> | ug/dl  | 4.5-12.0 |
| THYROID_STIMULATING_HORMONE | C.L.I.A | <input type="text" value="2.5"/> | uIU/ml | 0.30-5.5 |

*Technology:*

T3-Competitive Chemi Luminescent Immuno Assay

T4-Competitive Chemi Luminescent Immuno Assay

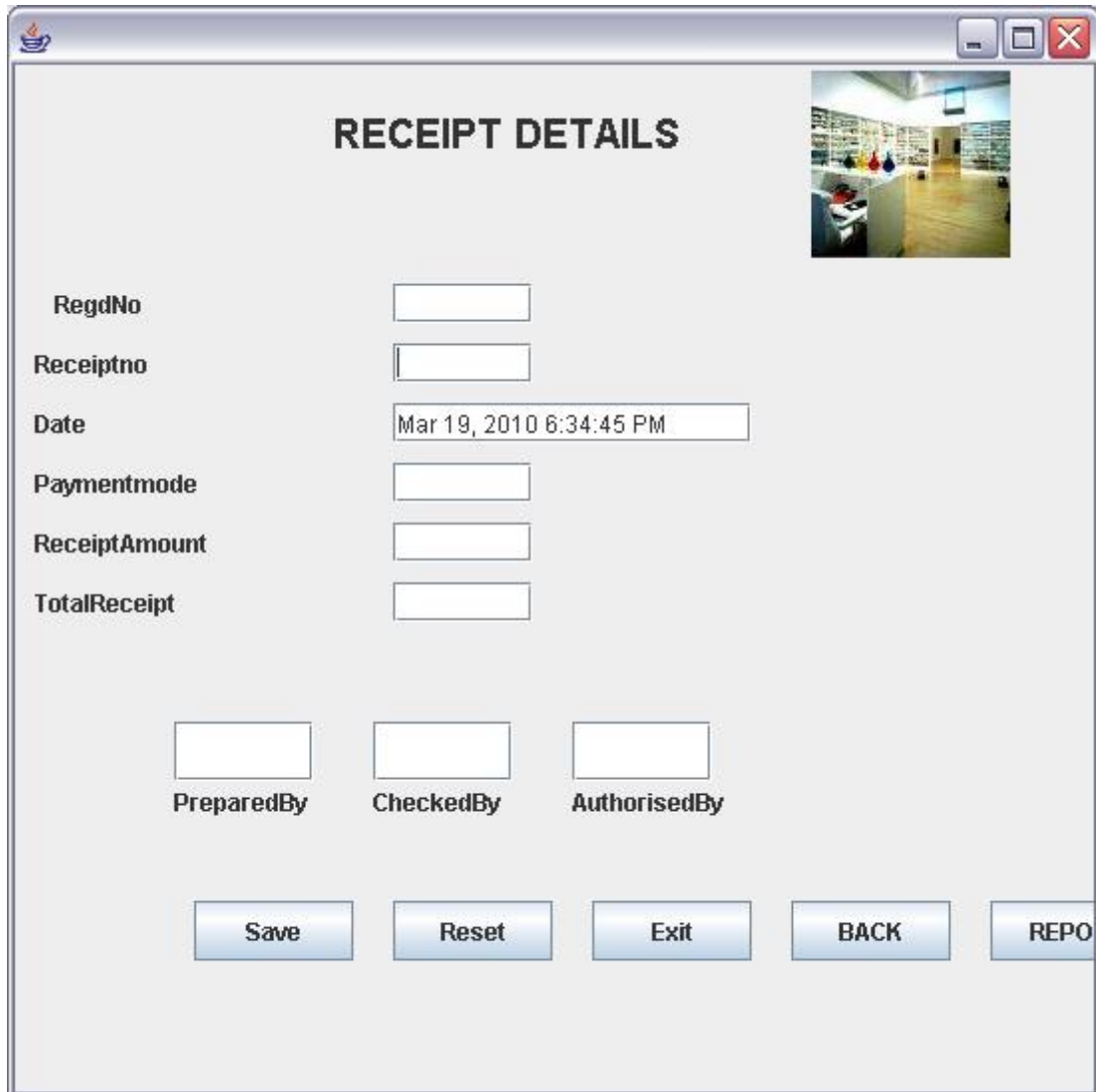
TSH-Ultra Sensitive Chemi Luminescent Immuno Assay

Dr.Mahesh & ...

Message

 RECORDS SAVED SUCCEESUFLY!!!!!!

***Input:-***



**RECEIPT DETAILS**



RegdNo

Receiptno

Date

Paymentmode

ReceiptAmount

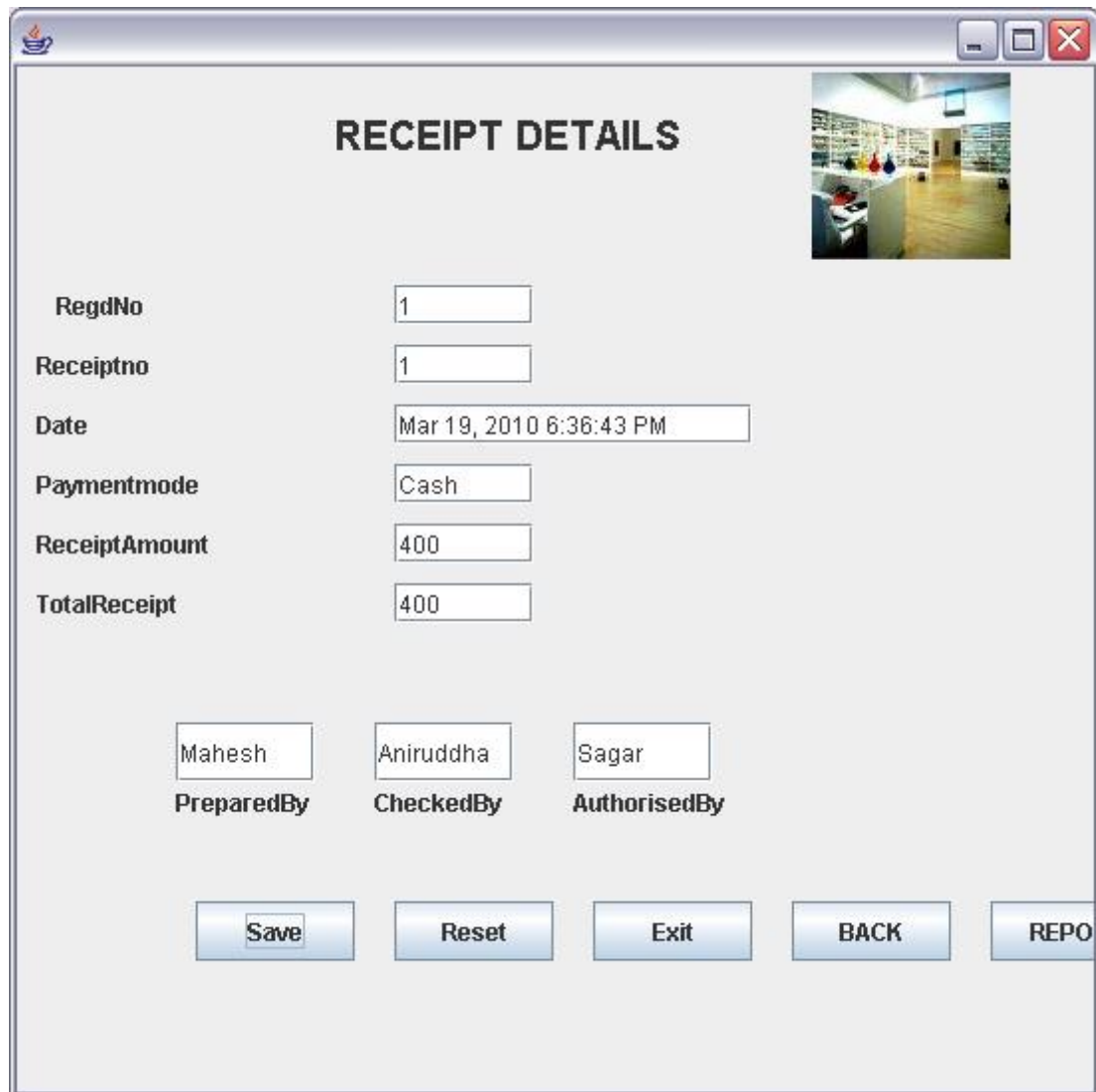
TotalReceipt

PreparedBy

CheckedBy

AuthorisedBy

## Output:-



**RECEIPT DETAILS**

RegdNo: 1

Receiptno: 1

Date: Mar 19, 2010 6:36:43 PM

Paymentmode: Cash

ReceiptAmount: 400

TotalReceipt: 400

PreparedBy: Mahesh

CheckedBy: Aniruddha

AuthorisedBy: Sagar

Buttons: Save, Reset, Exit, BACK, REPO

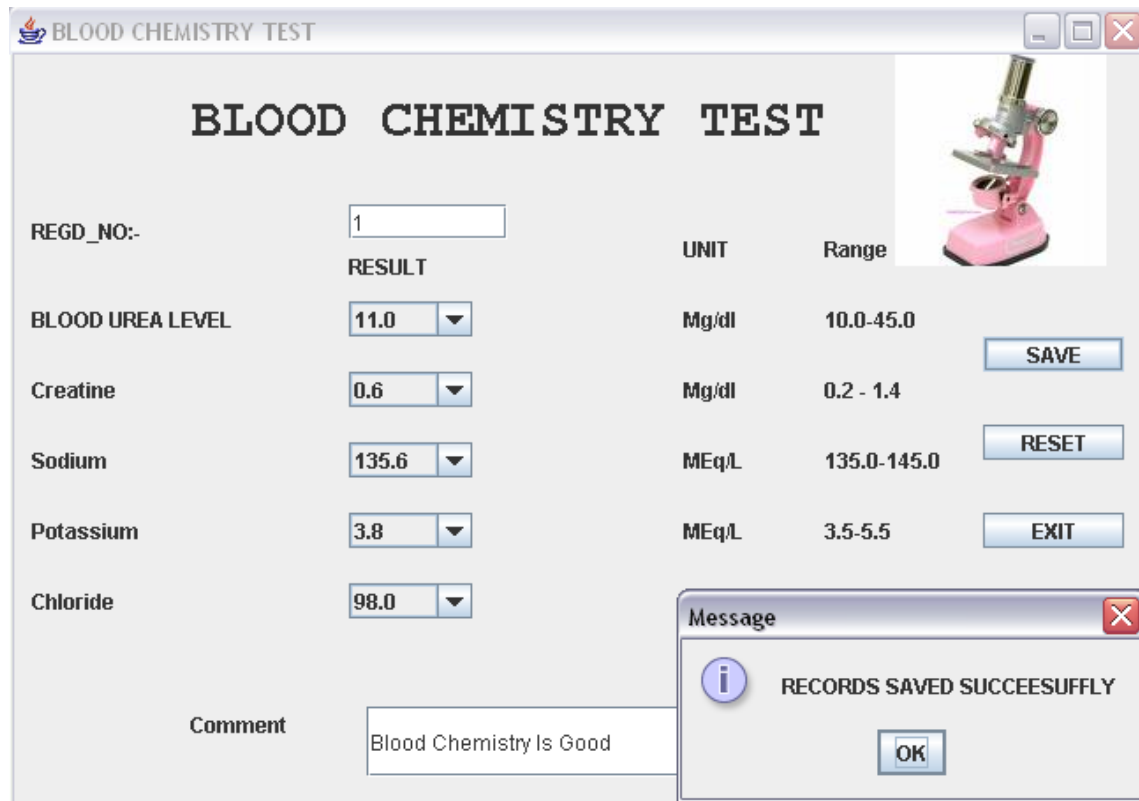
***Input:-***



## BLOOD CHEMISTRY TEST

|                  |                                    |       |             |
|------------------|------------------------------------|-------|-------------|
| REGD_NO:-        | <input type="text"/>               |       |             |
|                  | RESULT                             | UNIT  | Range       |
| BLOOD UREA LEVEL | <input type="text" value="0.2"/>   | Mg/dl | 10.0-45.0   |
| Creatine         | <input type="text" value="0.2"/>   | Mg/dl | 0.2 - 1.4   |
| Sodium           | <input type="text" value="135.0"/> | MEq/L | 135.0-145.0 |
| Potassium        | <input type="text" value="3.5"/>   | MEq/L | 3.5-5.5     |
| Chloride         | <input type="text" value="97.0"/>  | MEq/L | 97.0-111.0  |
| Comment          | <input type="text"/>               |       |             |

## Output:-

The image shows a graphical user interface for a "BLOOD CHEMISTRY TEST" application. The window has a title bar with the text "BLOOD CHEMISTRY TEST" and standard Windows window controls (minimize, maximize, close). Inside the window, the title "BLOOD CHEMISTRY TEST" is displayed in a large, bold, serif font. To the right of the title is a small image of a pink microscope. Below the title, there is a form with several input fields and buttons. The form is organized into two main columns. The left column contains labels for "REGD\_NO:-", "BLOOD UREA LEVEL", "Creatine", "Sodium", "Potassium", and "Chloride". The right column contains labels for "UNIT" and "Range". Between these columns are input fields for "RESULT" and "Comment". The "RESULT" field is a dropdown menu with values "11.0", "0.6", "135.6", "3.8", and "98.0". The "UNIT" field has values "Mg/dl", "Mg/dl", "MEq/L", and "MEq/L". The "Range" field has values "10.0-45.0", "0.2 - 1.4", "135.0-145.0", and "3.5-5.5". There are three buttons on the right side: "SAVE", "RESET", and "EXIT". A "Message" dialog box is open in the bottom right corner, displaying the text "RECORDS SAVED SUCCEESUFFLY" (note the typo) and an "OK" button. The "Comment" field contains the text "Blood Chemistry Is Good".

| REGD_NO:-        | RESULT | UNIT  | Range       |
|------------------|--------|-------|-------------|
| 1                | 11.0   | Mg/dl | 10.0-45.0   |
| BLOOD UREA LEVEL | 0.6    | Mg/dl | 0.2 - 1.4   |
| Creatine         | 135.6  | MEq/L | 135.0-145.0 |
| Sodium           | 3.8    | MEq/L | 3.5-5.5     |
| Potassium        | 98.0   |       |             |
| Chloride         |        |       |             |

Comment: Blood Chemistry Is Good

Message: RECORDS SAVED SUCCEESUFFLY

***Input:-***

| BLOOD SUGAR FASTING TEST |                      |      |        |
|--------------------------|----------------------|------|--------|
| RegdNo                   | RESULT               | UNIT | RANGE  |
| BLOOD SUGAR LEVEL        | <input type="text"/> | Mg % | 70-120 |
| URINE SUGAR              | PRESENT              |      |        |
| URINE ACETONE            | PRESENT              |      |        |
| comment                  | <input type="text"/> |      |        |

SAVE    RESET    EXIT    BACK    REP

## Output:-

**BLOOD SUGAR FASTING TEST**

RegdNo

|                   | RESULT                               | UNIT | RANGE  |
|-------------------|--------------------------------------|------|--------|
| BLOOD SUGAR LEVEL | <input type="text" value="71.0"/>    | Mg % | 70-120 |
| URINE SUGAR       | <input type="text" value="PRESENT"/> |      |        |
| URINE ACETONE     | <input type="text" value="PRESENT"/> |      |        |

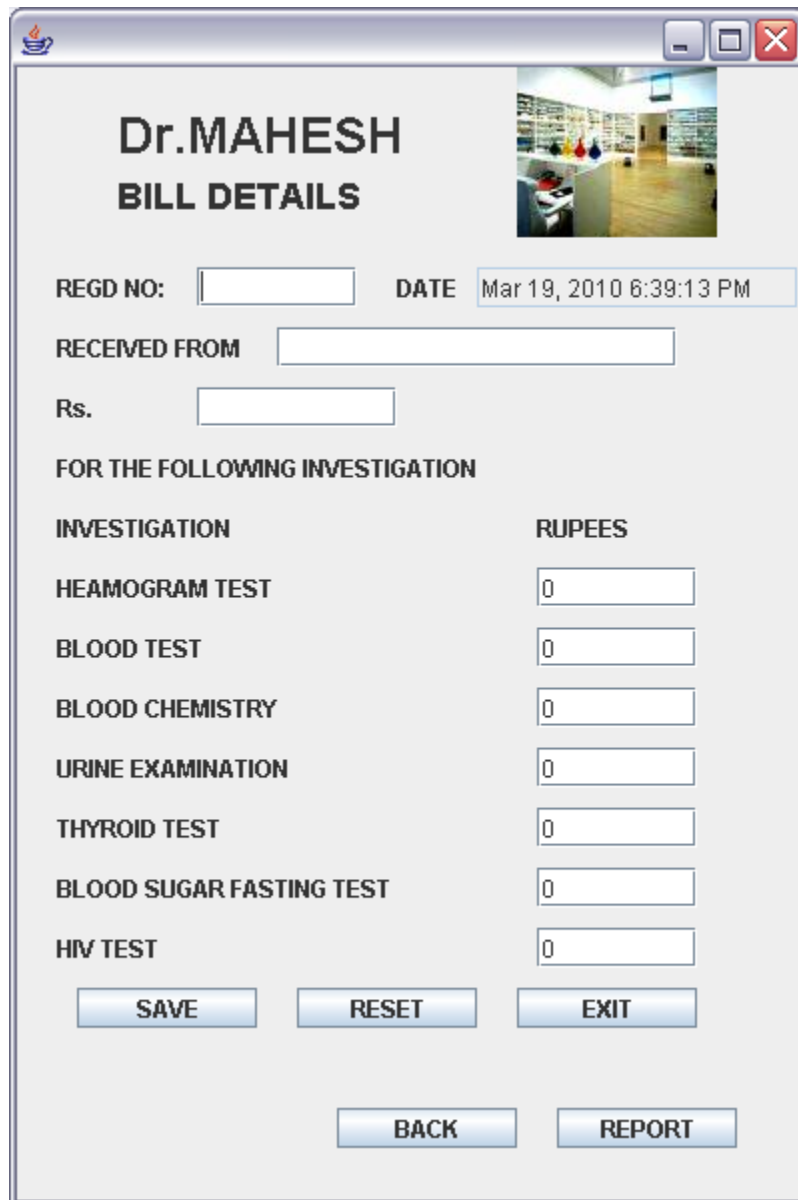
comment

**Message**

RECORDS SAVED SUCCEESUFFLY



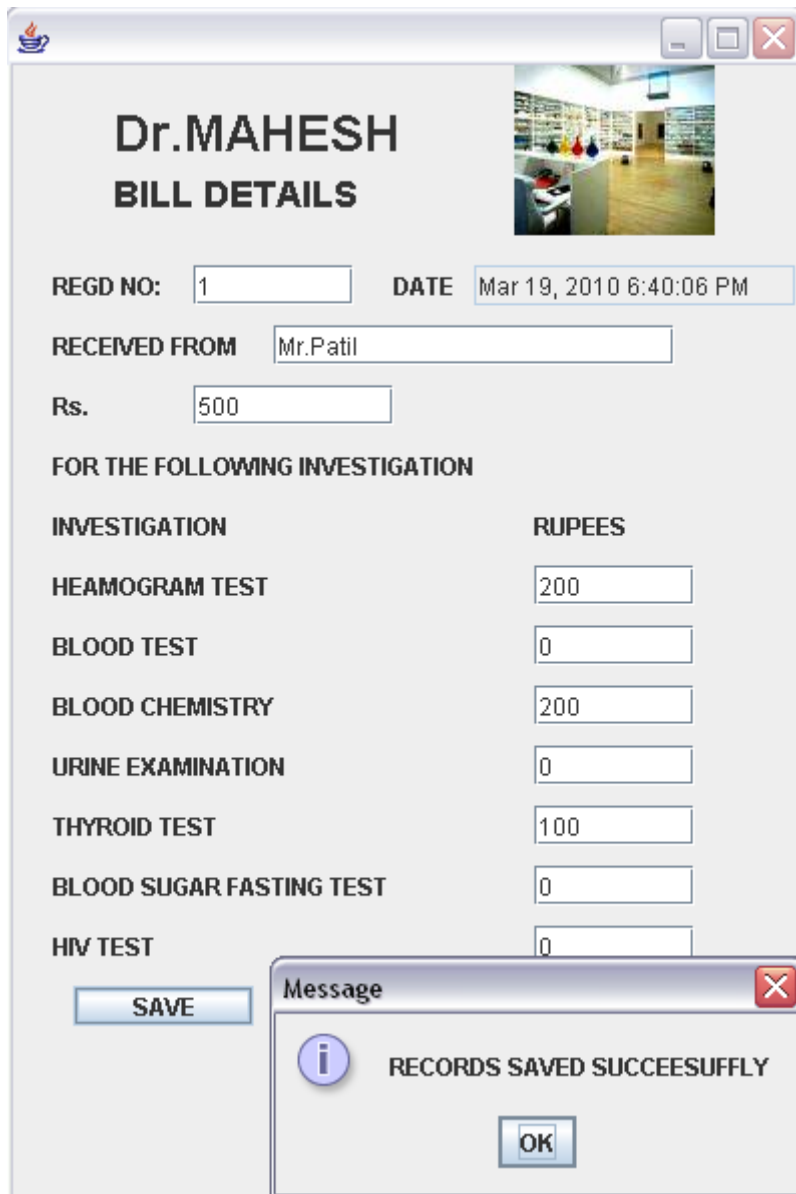
## ***Input:-***



The screenshot shows a software window titled "Dr. MAHESH BILL DETAILS". It contains several input fields and buttons for entering bill information. At the top right is a small image of a pharmacy interior. The form includes fields for "REGD NO:", "DATE" (pre-filled with "Mar 19, 2010 6:39:13 PM"), "RECEIVED FROM", and "Rs.". Below these is a section for "FOR THE FOLLOWING INVESTIGATION" with a table of tests and their costs in rupees. At the bottom are buttons for "SAVE", "RESET", "EXIT", "BACK", and "REPORT".

| INVESTIGATION            | RUPEES                         |
|--------------------------|--------------------------------|
| HEAMOGRAM TEST           | <input type="text" value="0"/> |
| BLOOD TEST               | <input type="text" value="0"/> |
| BLOOD CHEMISTRY          | <input type="text" value="0"/> |
| URINE EXAMINATION        | <input type="text" value="0"/> |
| THYROID TEST             | <input type="text" value="0"/> |
| BLOOD SUGAR FASTING TEST | <input type="text" value="0"/> |
| HIV TEST                 | <input type="text" value="0"/> |

## Output:-



**Dr.MAHESH**  
**BILL DETAILS**

REGD NO:  DATE:

RECEIVED FROM:

Rs.

FOR THE FOLLOWING INVESTIGATION

| INVESTIGATION            | RUPEES                           |
|--------------------------|----------------------------------|
| HEAMOGRAM TEST           | <input type="text" value="200"/> |
| BLOOD TEST               | <input type="text" value="0"/>   |
| BLOOD CHEMISTRY          | <input type="text" value="200"/> |
| URINE EXAMINATION        | <input type="text" value="0"/>   |
| THYROID TEST             | <input type="text" value="100"/> |
| BLOOD SUGAR FASTING TEST | <input type="text" value="0"/>   |
| HIV TEST                 | <input type="text" value="0"/>   |

Message

 RECORDS SAVED SUCCEESUFLY

## Advantages:-

How many patients present?

➤ How many reports are open from an individual patients ?

➤ Accessed information according to Bill modes

a) How many test occurs.

b) How many Patients.

c) How many reports.

d) How many times admin present.

➤ Access information through report.

So, accessing all above information manually is very difficult to admin at any time.

One of the best solutions is to maintain all client information through computer. By using Computer, Admin can access all information whatever he wants at any time also with great accuracy.

## Limitations:-

- System is not multi-user. It can be use by one person at a time.
- We cannot maintain information for long duration.
- In the existing system they have to maintain information in register books.
- The Existing system is time consuming.
- More human errors.
- Searching a record is a tedious task
- Existing system take more time for generating result.
- In Existing there is a possibility of loss of data.

## Future Enhancement

- ❖ The system is capable of creating report which is required by management. There can be more report & queries which may enhance the entire system.
- ❖ More Graphical user interface can be added. Security can be improved as professional security.
- ❖ Access does not provide much Database security so backend should change to MySQL or Oracle.

## **Bibliography**

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